

count digit

Problem statement

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You are given a number '*n*'.

Find the number of digits of '*n*' that evenly divide '*n*'.

Note:

A digit evenly divides '*n*' if it leaves no remainder when dividing '*n*'.

Example:

Input: '*n*' = 336

Output: 3

Explanation:

336 is divisible by both '3' and '6'. Since '3' occurs twice it is counted two times.

question understanding:

- A digit is said to evenly divide 'n' if, when 'n' is divided by that digit, there is no remainder.

examples:

m = 336

n = 336

% 10 =

6

last_digit



- check if **m % last_digit == 0**, then that is evenl divide.

$$m = 336 \% 6 = 0$$

$n = 336$



$n/10$

$$n = 33 \% 10 = 3$$

- check if $m \% \text{last_digit} == 0$, then that is even divide.

$$m = 336 \% 3 = 0$$

$n = 33$



$n/10$

$n = 3$

$\% 10 = 3$

- check if **$m \% \text{last_digit} == 0$** , then that is evenl divide.

$m = 336$

$\% 3 = 0$

$n = 3$

$n/10$

$n = 0$

stop

pseudocode:

C++ (g++ 5.4) ▾

```
1  int countDigits(int n){
2      int m = n;
3
4      int cnt = 0;
5
6      while (n) {
7          int last_digit = n % 10;
8
9          if (last_digit != 0) {
10
11              if (m % last_digit == 0) {
12                  cnt++;
13              }
14          }
15
16          n = n / 10;
17      }
18      return cnt;
19 }
```

code in c++

Sample test case Custom test case

- ✓ Test case 12 >
- ✓ Test case 13 >
- ✓ Test case 14 >
- ✓ Test case 15 >

▶ Run



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